



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CONVERSATION-STATE BASED APPROACH

Assignment on

*Assignment Report submitted in partial fulfilment of the requirements for the
Course of*

DSA0308 – Natural Language Processing

to the award of the degree of

BACHELOR OF Engineering

IN

**COMPUTER SCIENCE ENGINEERING WITH ARTIFICIAL
INTELLIGENCE**

Submitted by

AB DHANYASHREE(192472002)

Under the Supervision of

Dr.Kumaragurubaran T

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

SEPTEMBER 2026

CONVERSATION-STATE BASED APPROACH

1. Problem Statement and Problem Formulation

Problem Statement

An e-commerce chatbot may receive information about the same order over several conversational turns. A customer does not always repeat the complete information in every message. Instead, they may use expressions such as *it*, *this order*, *that item*, or *there* to refer to information already mentioned.

A chatbot that processes each message independently may lose this information and assign an incorrect intent. For example, in the given interaction, the customer first mentions a laptop order that has not arrived and later says:

“I don't want to cancel it. I just want to know when it will arrive.”

The meaning of “**it**” depends on the previous conversation. Similarly, the occurrence of *cancel* should not cause the chatbot to classify the message as Order Cancellation because the customer explicitly rejects cancellation.

The proposed solution therefore maintains a structured **conversation state** that is updated after every user message. This state stores entities, events, delivery status, temporal information, references, rejected actions and the current intent. The assignment requires conversational context and reference resolution to be handled explicitly.

Problem Formulation

Input:

Multiple customer messages belonging to one e-commerce conversation.

Internal Representation:

A continuously updated conversation state.

Output:

- Current intent
- Relevant entities
- Resolved references
- Conversation status
- Intent confidence
- Retrieved information
- Generated response

- Tamil translation

The possible intent classes are Order Tracking, Order Cancellation, Product Information, Return Request, Refund Request, Payment Problem, Delivery Information, Account Problem, Complaint and General Inquiry.

2. Objective and Expected Outcomes

Objectives

The main objective is to design a chatbot that understands the **current user intention using both the current message and previously stored conversation information.**

The specific objectives are:

1. Process each conversational turn.
2. Extract entities from every message.
3. Maintain an evolving conversation state.
4. Perform morphological processing.
5. Generate n-gram representations.
6. Perform POS tagging.
7. Analyze sentence structure using CFG.
8. Detect negated actions.
9. Build semantic representations.
10. Resolve references using previous turns.
11. Maintain entity chains.
12. Detect discourse relationships.
13. Determine the current intent from the complete state.
14. Retrieve information relevant to the intent.
15. Generate an appropriate response.
16. Translate the response into Tamil.

These objectives incorporate the course concepts specified in the assignment.

Expected Outcomes

For the given conversation, the state-based system should maintain:

Product = laptop
Order = existing customer order
Order Status = not received
Time = three days
Delivery Issue = delayed
Cancellation = rejected
Current Reference = existing order
Current Intent = order tracking
The final intent should therefore be:

Order Tracking / Delivery Information

3. Requirements, Constraints and Assumptions

Requirements

Functional Requirements

The chatbot must:

- accept multiple conversational turns;
- tokenize and normalize text;
- perform stemming;
- generate n-grams;
- perform POS tagging;
- construct syntactic representations;
- detect negation;
- perform semantic interpretation;
- identify entities;
- resolve references;
- maintain entity chains;
- analyze discourse;
- update the conversation state;
- determine the current intent;

- retrieve appropriate information;
- generate a response;
- translate the response.

The assignment specifically requires reference resolution for expressions such as *it*, *this* and *that*, along with conversational context maintenance.

State Requirements

The conversation state should be capable of storing:

ConversationState

```

├── customer
├── products
├── orders
├── order_status
├── delivery_status
├── temporal_information
├── references
├── negated_actions
├── discourse_relations
└── current_intent

```

Constraints

The system should not process every message as an isolated input.

It must:

- retain relevant previous information;
- resolve references from conversational context;
- explicitly identify negated actions;
- distinguish an entity from an intent;
- avoid replacing previously established facts without evidence;
- provide intermediate processing results;
- avoid generating responses inconsistent with the conversation.

These requirements are consistent with the technical constraints in the assignment.

Assumptions

1. Conversation turns arrive sequentially.
2. The system maintains state during a session.
3. The most recent compatible entity is generally preferred for resolving simple references.
4. Customer messages contain sufficient information for identifying the relevant intent.
5. A predefined collection of e-commerce intents exists.
6. The chatbot has access to a small information-retrieval store.
7. Translation is performed after response generation.

4. Application of Relevant Course Knowledge / Concepts

Unit I – Morphological Analysis

Each message is tokenized and stemmed before higher-level processing.

For example:

Word	Stemmed Form
ordered	order
arrived	arriv
delivered	deliv
cancelling	cancel
tracking	track

This allows different word forms to be associated with similar concepts.

Regular expressions can also identify structured information such as order numbers and dates.

Unit II – N-Grams and POS

N-grams are used to preserve local word relationships.

For:

“I don't want to cancel it.”

Possible features include:

Unigrams

want

cancel

Bigrams

want cancel

cancel it

Trigrams

don't want cancel

want cancel it

The n-gram containing the negation context is more informative than the isolated word *cancel*.

POS tagging is used to distinguish actions, objects and other grammatical components.

Unit III – CFG and Negation

A simplified grammar can represent the structure of a customer statement:

$S \rightarrow NP VP$

$VP \rightarrow Aux VP$

$VP \rightarrow V NP$

$NP \rightarrow Det N$

The system separately detects negation.

For:

“I don't want to cancel it.”

the state records:

Action = cancellation

Negated = TRUE

This prevents the cancellation action from becoming the current customer goal.

Unit IV – Semantic Analysis and WSD

The semantic representation describes events and relationships.

For example:

Customer ordered Laptop

Order has not arrived

Customer requests arrival information

Customer rejects cancellation

A simplified FOPC representation is:

Ordered(Customer, Laptop)

NotArrived(Order)

Wants(Customer, ArrivalInformation)

NotWants(Customer, Cancellation)

WSD helps interpret terms according to the e-commerce context.

For example, *order* is interpreted as a purchase transaction rather than another possible meaning.

Unit V – Discourse, Reference and Context

This is the **central component of Version 3**.

The system creates an entity chain:

laptop → order → it → existing order

The word “**it**” is therefore not interpreted independently.

Similarly:

- **but** → introduces contrast;
- **still** → indicates an unresolved condition;
- **just** → restricts the customer's requirement.

The assignment explicitly identifies reference resolution, entity chains, discourse relations and coherence as Unit V concepts.

5. Design / Proposed Solution / Methodology

The proposed architecture is organized around a **Conversation State Manager**.

Customer Turn



Preprocessing



Morphological Analysis



N-Gram + POS Analysis



CFG + Negation



Semantic Analysis



Entity Extraction



Reference Resolution



Discourse Analysis



UPDATE CONVERSATION STATE



Current Intent Identification



Information Retrieval



Response Generation



Tamil Translation



Next Conversation Turn

The state is updated after every user message.

Conversation State Example

After the first message:

“I ordered a laptop three days ago, but it still hasn't arrived.”

the state becomes:

ConversationState

- |— product = laptop
- |— order_exists = TRUE
- |— order_age = three days
- |— arrival_status = not_arrived
- |— delivery_problem = TRUE
- |— probable_intent = order_tracking

After the second message:

“I don't want to cancel it. I just want to know when it will arrive.”

the state is updated:

ConversationState

- |— product = laptop
- |— order_exists = TRUE
- |— order_age = three days
- |— arrival_status = not_arrived
- |— cancellation = REJECTED
- |— current_reference = order
- |— requested_information = arrival_time
- |— current_intent = order_tracking

This state representation is the major distinction of this solution.

6. Algorithm / Pseudocode

BEGIN

Initialize ConversationState

FOR each user_message:

 normalized_text ← preprocess(user_message)

 tokens ← tokenize(normalized_text)

 stems ← perform_stemming(tokens)

 ngrams ← generate_ngrams(tokens, 1, 3)

 pos_tags ← POS_tag(tokens)

 parse_tree ← CFG_parse(user_message)

 negation_info ← detect_negation(user_message)

 semantic_info ← semantic_analysis(user_message)

 entities ← extract_entities(user_message)

```
references ← find_references(user_message)
```

FOR each reference:

```
candidate_entities ← obtain_entities_from_state()
```

```
resolved_entity ← select_best_contextual_match(  
    reference,  
    candidate_entities)
```

END FOR

```
discourse_info ← analyze_discourse(user_message)
```

update ConversationState with:

entities

resolved references

temporal information

order information

delivery status

negated actions

discourse relations

```
candidate_intents ← generate_possible_intents(  
    current_message,  
    ConversationState)
```

FOR each candidate intent:

calculate linguistic evidence

calculate state/context evidence

IF intent is explicitly negated:

 reduce intent score

IF intent agrees with stored state:

 increase intent score

END FOR

current_intent ← highest_valid_score

retrieved_data ← retrieve_information(
 current_intent,
 ConversationState)

response ← generate_response(
 current_intent,
 retrieved_data)

translated_response ← translate_to_tamil(response)

display state

display resolved references

display intent

display response

END FOR

END

The algorithm contains the functions, data structure, loops, conditions and scoring required by the assignment's algorithm-design expectations.

7. Implementation / Source Code and Environment / Tools Used

Environment

Component	Tool
Programming Language	Python
NLP Library	NLTK
Regex Processing	Python re
Tokenization	NLTK
Stemming	PorterStemmer
POS Tagging	NLTK
CFG Parsing	NLTK
State Management	Python dictionaries/classes
Intent Processing	Python
Information Retrieval	Dictionary-based knowledge base
Translation	Suitable translation library/API

Conversation State Implementation

A Python class can represent the state:

```
class ConversationState:
```

```

def __init__(self):
    self.product = None
    self.order = None
    self.order_status = None
    self.delivery_status = None
    self.time = None
    self.references = {}
    self.negated_actions = []
    self.discourse = []
    self.intent = None

```

After every message, the state is updated rather than recreated.

For example:

```

state.product = "laptop"
state.order_status = "existing"
state.delivery_status = "not_received"
state.time = "three days"
state.negated_actions.append("cancel")
state.intent = "order_tracking"

```

The important feature is that the second message can access the information extracted from the first message.

8. Test Cases and Expected / Actual Results

Test Case	Conversation	Expected Intent
TC1	“Where is my order?”	Order Tracking
TC2	“Cancel my order.”	Order Cancellation
TC3	“My package hasn't arrived.”	Order Tracking
TC4	“Can I return this item?”	Return Request

Test Case	Conversation	Expected Intent
TC5	"I paid but the payment failed."	Payment Problem
TC6	"I don't want to cancel it. When will it arrive?" after discussing an order	Order Tracking
TC7	"What is its delivery status?" after mentioning an order	Order Tracking

Multi-Turn Test

Turn 1

"I ordered a laptop three days ago."

State:

product = laptop

order = existing

time = three days

Turn 2

"It hasn't arrived yet."

State becomes:

product = laptop

order = existing

delivery_status = not_received

reference("it") = order

Turn 3

"I don't want to cancel it."

State becomes:

reference("it") = order

cancellation = rejected

Turn 4

"I just want to know when it will arrive."

Final state:

requested_information = arrival_time

intent = order_tracking

Thus, the system uses **all four turns** rather than interpreting the final message independently.

9. Execution Screenshots / Outputs

The actual implementation should capture screenshots for the following.

Screenshot 1 – Preprocessing/tokenization output.

```
1. PREPROCESSING / TOKENIZATION
-----
['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', '.', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', ',', 'but', 'the', 'battery', 'is', 'disappointing', '.', 'it', 'does', 'not', 'last', 'long', ',', 'and', 'i', 'am', 'unhappy', 'with', 'it', '.', 'however', ',', 'the', 'service', 'team', 'replaced', 'it', 'quickly', ',', 'which', 'was', 'impressive', '.', 'overall', ',', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']
```

User:

I ordered a laptop three days ago,

but it still hasn't arrived.

Screenshot 2 – N-gram and POS-tagging output screenshot.

```
3. PHRASES
-----
[['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', ',', 'but', 'the', 'battery', 'is', 'disappointing', 'it', 'does', 'not', 'last', 'long', 'and', 'i', 'am', 'unhappy', 'with', 'it', 'however', 'the', 'service', 'team', 'replaced', 'it', 'quickly', 'which', 'was', 'impressive', 'overall', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']]

4. BIGRAMS
-----
[['i', 'was', 'was', 'expecting', 'expecting', 'the', 'the', 'phone', 'phone', 'to', 'to', 'be', 'be', 'amazing', 'amazing', ',', ',', 'the', 'the', 'display', 'display', 'is', 'is', 'bright', 'bright', 'and', 'and', 'the', 'the', 'camera', 'camera', 'takes', 'takes', 'excellent', 'excellent', 'pictures', 'pictures', ',', ',', 'but', 'but', 'the', 'the', 'battery', 'battery', 'is', 'is', 'disappointing', 'disappointing', 'it', 'it', 'does', 'does', 'not', 'not', 'last', 'last', 'long', 'long', 'and', 'and', 'i', 'i', 'am', 'am', 'unhappy', 'unhappy', 'with', 'with', 'it', 'it', 'however', 'however', 'the', 'the', 'service', 'service', 'team', 'team', 'replaced', 'replaced', 'it', 'it', 'quickly', 'quickly', 'which', 'which', 'was', 'was', 'impressive', 'impressive', 'overall', 'overall', 'i', 'i', 'am', 'am', 'satisfied', 'satisfied', 'with', 'with', 'the', 'the', 'phone', 'phone', '.']]

5. TRIGRAMS
-----
[['i', 'was', 'expecting', 'was', 'expecting', 'the', 'the', 'phone', 'to', 'phone', 'to', 'be', 'to', 'be', 'amazing', 'be', 'amazing', ',', ',', 'amazing', 'the', 'the', 'display', 'display', 'is', 'is', 'display', 'is', 'bright', 'bright', 'and', 'and', 'the', 'the', 'camera', 'camera', 'takes', 'takes', 'excellent', 'excellent', 'pictures', 'pictures', ',', ',', 'but', 'but', 'the', 'the', 'battery', 'battery', 'is', 'is', 'disappointing', 'disappointing', 'it', 'it', 'does', 'does', 'not', 'not', 'last', 'last', 'long', 'long', 'and', 'and', 'i', 'i', 'am', 'am', 'unhappy', 'unhappy', 'with', 'with', 'it', 'it', 'however', 'however', 'the', 'the', 'service', 'service', 'team', 'team', 'replaced', 'replaced', 'it', 'it', 'quickly', 'quickly', 'which', 'which', 'was', 'was', 'impressive', 'impressive', 'overall', 'overall', 'i', 'i', 'am', 'am', 'satisfied', 'satisfied', 'with', 'with', 'the', 'the', 'phone', 'phone', '.']]

6. QUADGRAMS
-----
[['i', 'was', 'expecting', 'was', 'expecting', 'the', 'the', 'phone', 'to', 'phone', 'to', 'be', 'to', 'be', 'amazing', 'be', 'amazing', ',', ',', 'amazing', 'the', 'the', 'display', 'display', 'is', 'is', 'display', 'is', 'bright', 'bright', 'and', 'and', 'the', 'the', 'camera', 'camera', 'takes', 'takes', 'excellent', 'excellent', 'pictures', 'pictures', ',', ',', 'but', 'but', 'the', 'the', 'battery', 'battery', 'is', 'is', 'disappointing', 'disappointing', 'it', 'it', 'does', 'does', 'not', 'not', 'last', 'last', 'long', 'long', 'and', 'and', 'i', 'i', 'am', 'am', 'unhappy', 'unhappy', 'with', 'with', 'it', 'it', 'however', 'however', 'the', 'the', 'service', 'service', 'team', 'team', 'replaced', 'replaced', 'it', 'it', 'quickly', 'quickly', 'which', 'which', 'was', 'was', 'impressive', 'impressive', 'overall', 'overall', 'i', 'i', 'am', 'am', 'satisfied', 'satisfied', 'with', 'with', 'the', 'the', 'phone', 'phone', '.']]
```

Screenshot 3 – Reference Resolution

```
10. WORD SENSE DISAMBIGUATION
-----

Word: service
Possible senses: customer support, software service
Selected sense: customer support
Reason: The context mentions a service team replacing the phone.

Word: display
Possible senses: phone screen, public presentation
Selected sense: phone screen
Reason: The word occurs in the context of a phone.

Word: battery
Possible senses: phone battery, generic battery
Selected sense: phone battery
Reason: The review is explicitly about a phone.

11. REFERENCE RESOLUTION
-----
It in 'It does not last long' -> Battery
it in 'I am unhappy with it' -> Phone
it in 'replaced it quickly' -> Phone
which in 'which was impressive' -> Service/replacement action
```

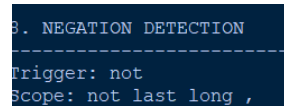
Reference:

"it"

Resolved To:

Existing Order

Screenshot 4 – Negation Analysis



```
B. NEGATION DETECTION
-----
Trigger: not
Scope: not last long ,
```

Detected Action:

Cancellation

Negation:

TRUE

Interpretation:

Customer rejects cancellation

Screenshot 5 – Final State

Conversation State

Product : Laptop

Order Status : Existing

Delivery Status : Not Received

Cancellation : Rejected

Current Reference : Order

Requested Information: Arrival Time

Current Intent : Order Tracking

Screenshot 6 – Response

Final Intent:

Order Tracking

Response:

Please provide your order number so I can check
the current delivery status of your laptop.

Tamil Response:

இந்த மதிப்பாய்வு ஒட்டுமொத்தமாக நேர்மறையானதாக வகைப்படுத்தப்படுகிறது. இரை மற்றும் கேமரா குறித்து நேர்மறையான கருத்துகள் உள்ளன. அதே நேரத்தில், பேட்டரி ஏமாற்றமளிப்பதாகவும் நீண்ட நேரம் நடிக்கவில்லை என்றும் கூறப்படுவதால் அது எதிர்மறையான அம்சமாக உள்ளது. சேவை அணியினர் தொலைபேசியை விரைவாக மாற்றியதால் சேவை அனுபவம் நேர்மறையாக உள்ளது. இறுதியாக, பயனர் தொலைபேசியில் திருப்தியாக இருப்பதாகக் கூறுகிறார். எனவே, ஒட்டுமொத்த மதிப்பீடு நேர்மறையாகும்.

The assignment requires evidence of intermediate NLP representations, final classification and generated chatbot output.

10. Results and Validation

Morphological Analysis

Word	Processed Form
ordered	order
arrived	arriv
cancelling	cancel
delivery	deliver
tracking	track

Conversation-State Validation

State Component	Value	Source
Product	Laptop	First turn
Order	Existing	First turn
Time	Three days	First turn
Delivery Status	Not received	First/second turn
Reference	“it” → order	Previous context
Cancellation	Rejected	Third turn
Requested Information	Arrival time	Final turn
Intent	Order Tracking	Complete conversation

Intent Validation

The critical distinction is:

cancel



Potential cancellation signal

don't want to cancel



Cancellation rejected

want to know when it will arrive



Delivery/Tracking requirement

Therefore:

Final Intent: Order Tracking / Delivery Information

The assignment specifically requires the analysis to explain why *“I don't want to cancel it”* must not lead to Order Cancellation.

Validation Cases

Linguistic Feature	System Behaviour
Morphological variation	Normalizes related forms
N-gram dependency	Considers surrounding words
POS information	Provides grammatical evidence
Negation	Rejects negated intent
Reference	Resolves “it” from state
Ambiguity	Uses conversational context
Discourse	Interprets “but”, “still”, “just”
Multi-turn context	Retains previous information
Intent	Selects current user goal
Response	Generated according to state

These correspond to the validation areas required in the assignment.

11. Analysis, Comparison, Trade-offs and Justification

The major advantage of this approach is that it does not treat every message as an independent classification problem.

Consider:

“I ordered a laptop three days ago.”

At this point:

Intent = possibly Order Tracking

Then:

“It hasn't arrived.”

The reference “**it**” becomes meaningful because the state already contains the order.

Finally:

“I don't want to cancel it.”

The system already knows what *it* represents.

Importance of Reference Resolution

Without state:

"it" → unknown

With state:

"it" → existing order

This makes the chatbot much more capable of handling natural conversational behaviour.

Why “cancel” Does Not Become the Intent

The system stores:

negated_actions = ["cancel"]

Therefore:

cancel ≠ requested_intent

Instead:

requested_information = arrival time

and consequently:

intent = order tracking

Comparison

Approach	Main Advantage	Main Limitation
Keyword-based	Simple implementation	Cannot preserve context
Rule-based scoring	Explainable	Requires many manually designed rules
TF-IDF classifier	Learns statistical patterns	Limited conversation memory
Conversation-state model	Strong multi-turn understanding	State management is more complex
Proposed state-based approach	Context + linguistic processing	Requires careful state updates

Trade-off

The main trade-off is **complexity versus conversational reliability**.

Maintaining state requires additional memory and processing, but it substantially improves the system's ability to understand references and previous information.

This is particularly valuable in e-commerce because customers often refer back to an order without repeating its complete details.

12. Broader Considerations / SDG Relevance

The assignment maps the application to **SDG 9 – Industry, Innovation and Infrastructure**, with automated customer-service systems identified as an industry-relevant application.

Privacy

Conversation state may contain customer information such as products, order details and delivery information. The state should be protected and deleted appropriately after the session.

Bias

Reference resolution may perform differently depending on sentence construction and language style. Testing should include varied conversational expressions.

Incorrect State Updates

A major risk of this approach is storing an incorrect entity or reference. Once incorrect information enters the state, later responses may also become incorrect.

Accessibility

Tamil translation allows the system to provide customer support in a regional language.

Responsible AI

The chatbot should not perform irreversible actions merely because an intent was inferred. Actions such as cancellation should require explicit confirmation.

13. Conclusion, Limitations and Possible Improvements

Conclusion

A conversation-state-based e-commerce chatbot was designed to address the limitations of independent message classification. The system maintains information from previous turns and uses this information to interpret references, negation, entities and discourse.

The conversation state stores details such as the product, order status, delivery condition, temporal information, references and rejected actions.

For the given example, “**it**” is resolved to the customer's existing order, while the cancellation request is identified as negated. The customer's actual requirement is to determine when the order will arrive. Therefore, the final intent is **Order Tracking / Delivery Information**.

This demonstrates that maintaining conversational state can significantly improve the interpretation of multi-turn e-commerce interactions.

Limitations

- Incorrect entity extraction can affect later turns.
- Very long conversations can produce large states.
- Ambiguous references may have multiple possible antecedents.
- State management requires additional implementation effort.
- Complex discourse relations may require more advanced NLP models.
- The approach may struggle when customers suddenly change topics.

Possible Improvements

Future development could include:

- Transformer-based dialogue-state tracking.
- Advanced coreference resolution.
- Long-context conversational models.
- Automatic state-confidence estimation.
- Multilingual conversation-state tracking.
- Integration with real-time order-management systems.
- Human-agent escalation for uncertain states.

14. Individual Contribution of Group Members

The following is an example structure and should be modified according to the **actual contributions**.

Member Contribution

Member 1 Conversation-state architecture and problem formulation

Member 2 Preprocessing, stemming and POS processing

Member 3 Entity extraction and reference resolution

Member 4 Negation, discourse analysis and intent determination

Member 5 Response generation, Tamil translation, testing and documentation

15. References

The final report should cite the resources actually used during development. The assignment requires appropriate references for textbooks, research papers, NLP libraries, datasets, software/tools, websites and AI-assisted tools where applicable.

Suggested categories include:

1. NLP course textbook.
2. NLTK documentation.
3. Python documentation.
4. Research literature on dialogue-state tracking and conversational NLP.
5. Documentation for the selected translation library/API.
6. Dataset documentation, if a dataset is used.
7. Software and development tools used during implementation.
8. AI-assisted tools, if actually used.

16. One-Page Individual Reflection

Individual Reflection

While completing this assignment, I learned that a conversational chatbot cannot always understand a user's intention from the latest message alone. In real conversations, users often

depend on information mentioned in earlier turns. Because of this, an effective chatbot needs to retain previous conversational information and use it when interpreting new messages. Based on this idea, I developed the system using a structured conversation-state approach.

One of the main design decisions was to create a structured state that stores important information such as entities, order details, delivery status, time-related information, references, rejected or negated actions, and the current user intent. Rather than treating every message independently, the system continuously updates the conversation state as new messages arrive. This helped the system preserve relationships between different parts of the dialogue.

Reference resolution was another important NLP component used to determine the correct meaning of a user's message. For example, the meaning of the pronoun “it” cannot always be understood without considering the previous conversation. If “it” is correctly connected to an order discussed earlier, the chatbot can better identify the user's intention. Similarly, the statement “I don't want to cancel it” should be understood as refusing or rejecting cancellation instead of being mistakenly classified as a request to cancel.

This assignment also helped me understand how different NLP techniques can complement each other. Morphological processing helps normalize different word forms, while n-grams provide useful information about relationships between neighboring words. POS tagging helps identify the grammatical role of words, and semantic analysis contributes to understanding the meaning of the complete message. When these techniques are combined with discourse information and the current conversation state, the system can achieve a more reliable interpretation of user input.

A major challenge I experienced was determining how the conversation state should be updated after every new message. An incorrect state update can cause errors in the interpretation of subsequent messages. This taught me that the system needs proper confidence checking and should handle uncertain or ambiguous references carefully before modifying the existing conversation state.

The project is also related to **SDG 9 – Industry, Innovation and Infrastructure**, as conversational AI can be applied to automated customer-service and other digital support systems. The use of Tamil translation can further improve accessibility by allowing users who are more comfortable communicating in Tamil to interact with the chatbot more easily.

If I had more time to develop the project, I would improve the reference-resolution technique and implement a more sophisticated dialogue-state tracking approach. I would also evaluate the system using longer and more complex conversations, especially situations where the topic changes during the discussion or where several orders are mentioned at the same time.

Overall, this assignment gave me a practical understanding of how different NLP concepts can be combined to build a context-aware conversational system. It strengthened my knowledge of context handling, linguistic processing, algorithm development, and the practical application of NLP techniques in real-world systems.